

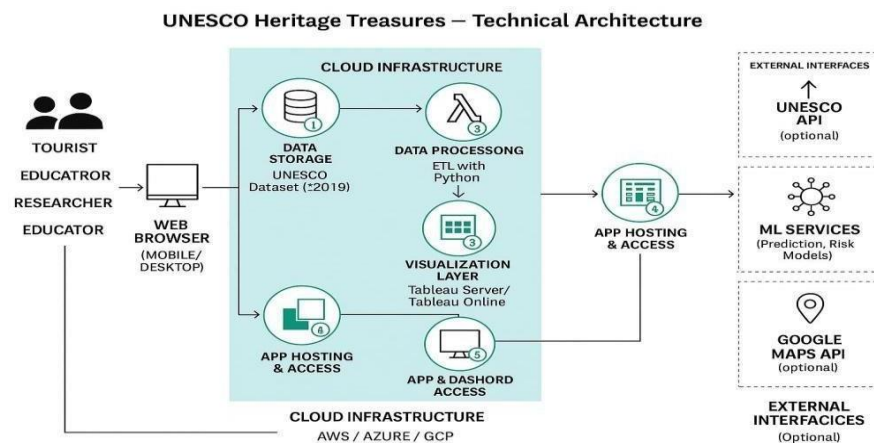
Project Design Phase-II Technology Stack (Architecture & Stack)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Example:



Guidelines:

- Show all key steps: data storage, processing, visualization, and access.
- Separate IBM Cloud components from user and external tools.
- Include external APIs like Google Maps or UNESCO API if used.
- Mark where raw and cleaned data is stored (e.g., IBM Cloud Storage).
- Label each layer: processing, visualization, user interface, etc.
- Add user types like tourists, researchers, or policymakers.
- Show machine learning if it's used or planned in the system.

Table-1: Technology Stack

S.no	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	Interface of Tableau
2.	Application Logic-1	Logic for a process in the application	tableau public
3.	Application Logic-2	Logic for a process in the application	tableau public
4.	Application Logic-3	Logic for a process in the application	tableau public
5.	Database	Data Type, Configurations etc.	My SQL
6.	Cloud Database	Database Service on Cloud	-
7.	File Storage	File storage requirements	-
8.	External API-1	Purpose of External API used in the application	-
9.	External API-2	Purpose of External API used in the application	-

10.	Machine Learning Model	Purpose of Machine Learning Model	-
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration	-

Table-2: Application Characteristics:

S.no	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	-
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Microservices)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

<https://c4model.com/> <https://developer.ibm.com/patterns/online-order-processing-system-duringpandemic/>

<https://www.ibm.com/cloud/architecture> <https://aws.amazon.com/architecture> <https://medium.com/theinternal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>